GRE Probability Formula and Practice Sheet

Permutations vs Combinations, The Slot Method, Complementary Probability & Overlapping Sets

1. Permutations vs Combinations: The Foundational Difference

Counting questions on the GRE boil down to one fundamental decision: DOES ORDER MATTER?

Counting Dimension	Permutation (Order Matters!)	Combination (Order Does NOT Matter!)
Guiding Question	Does changing the arrangement create a DIFFERENT outcome?	Does changing the arrangement result in the SAME group?
Classic Real-World Examples	Race podiums (1st, 2nd, 3rd), Passcodes, Officers (President, VP)	Committees, Selecting team members, Choosing ice cream toppings
Mathematical Formula	nPr = n! / (n - r)!	nCr = n! / [r! * (n - r)!]
Calculation Shortcut	The "Slot Method": Fill positions sequentially with decreasing choices	Calculate the Permutation first, then divide by r! to remove duplicates
Relative Quantity Result	Always LARGER (more arrangements possible)	Always SMALLER (divides out duplicate orderings)

Example: Selecting 3 committee members from 5 people is 5C3 = 10. Assigning them President, VP, and Secretary is 5P3 = 60.

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2. Core Probability Rules and Calculation Shortcuts

Master the fundamental rules of independent events, mutually exclusive outcomes, and complementary probability.

Probability Law / Rule	Mathematical Equation	Application Scenario	Calculation Shortcut
Basic Probability Definition	P(E) = (Favorable Outcomes) / (Total Possible Outcomes)	Any random uniform trial (dice, cards, coins)	Count total sample space first before counting favorable
Complementary Probability	P(Event Occurring) = 1 - P(Event NOT Occurring)	Crucial for questions containing "AT LEAST ONE"	P(At least 1 head in 4 coin flips) = 1 - P(Zero heads) = 1 - (1/16) = 15/16!
Independent Events (AND)	P(A and B) = P(A) * P(B)	Events that do not affect each other (rolling two dice)	Multiply probabilities directly
Dependent Events (Without Replacement)	P(A and B) = P(A) * P(B after A)	Drawing marbles or cards from a deck without replacement	Decrease numerator and denominator for the second draw
Mutually Exclusive (OR)	P(A or B) = P(A) + P(B) - P(A and B)	Events that can potentially happen at the same time	Subtract the overlap to prevent double counting

Golden GRE Shortcut: Whenever a probability question asks for "at least one", ALWAYS calculate 1 - P(none).

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3. Top 4 Counting and Probability Traps on the GRE

Common errors that trap test-takers on medium and hard Quant questions.

- Trap 1: Forgetting to Divide by Duplicates in Combinations. If selecting a group, you must divide by the factorial of group size.
- Trap 2: Double Counting in Overlapping Sets. In probability questions with overlaps, always use P(A) + P(B) - P(Both).
- Trap 3: Not Accounting for Replacement. Read carefully whether items are drawn "with replacement" or "without replacement".
- Trap 4: Over-Complicated Factorial Arithmetic. Always write out factorials and cancel terms before multiplying: 10! / 8! = 10 * 9 = 90.